CS353 - DATABASE SYSTEMS

Culinary Discovery & Meal Kit Marketplace

Project Proposal

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1. Problem Description

1.1 Overview of The System

Culinary Discovery & Meal Kit Marketplace is a web-based application that bridges between real-time users, that is people who cook, and real-time suppliers. The system operates with a Farm-To-Table philosophy. It offers recipes filtered according to users' dietary habits and preferences, scales the recipe according to the number of servings the user prefers, and allows them to purchase the required ingredients in specified quantities from suppliers. Unlike standard recipe sites, the application integrates a marketplace into this concept. Based on the selected recipe and number of servings, the system checks the supplier's stocks, even suggests alternative products when the desired product is not available, and creates a shopping cart with exact quantities. Not staying limited to individual use, the system also provides social interactions. It achieves this through kitchen challenges created from time to time and motivates users by providing appropriate rewards.

1.2 Why and How A Database Is Necessary

The use of a relational database in the design of this system is quite necessary because the system has to manage a large volume of interconnected and complex data structures. On the platform, it manages ingredient lists organized by recipes, preparation steps of recipes, stock levels of suppliers, pricing, reviews, challenges, user profiles separated both by category and individually, payments, and many other critical features like these. All these entities require efficient queries and therefore, must be connected to each other with logical relationships. Without a well and systematically designed relational database, it would not be possible to maintain the consistency of data in this system, enforce system rules, and effectively manage the flow of transactions.

1.3 Target Users and Their Roles

Home Cook: Home Cook is the main user of the platform. The role of this user in the system is as follows: searches for recipes based on dietary restrictions, preferences, and recipe contents; selects the meal they prefer to cook, and with the Shop This Meal feature, the ingredients of that meal are added to the cart directly as a whole based on the portion count determined by the home cook. Additionally, these users can also upload their own recipes to the system and can also engage in social interaction by participating in Kitchen Challenges. As Home Cooks apply the recipes of verified chefs, they earn royalty points for them while they themselves earn loyalty points.

Verified Chef: Unlike the recipes published by Home Cooks, the recipes of Verified Chefs are linked with Local Suppliers. The ingredients of the recipe can be directly sourced from Local Suppliers with the amounts specified in the recipe. There is a royalty metric that shows how popular their recipes are, meaning how much they have been purchased and cooked, and the reviews.

Local Supplier: These users are local food suppliers or sellers who manage their product inventories, stock quantities, pricing, and expiration dates through the system. When Home Cooks want to place an order with the Shop This Meal feature, the inventories of these suppliers are queried.

Administrator: This user is generally responsible for the management of the system. Has the authority to manage user accounts, permissions, verifications of chefs, and verifications of suppliers. Oversees the processing of certain recipes into certain categories and generates analyses.

2. System Requirements

2.1 Functional Requirements

1. The system must allow Home Cooks to search for and filter recipes by dietary restrictions, cooking time, ingredients, and chef rating.
2. The system must display the matching recipes with their titles, chef names, ratings, and cooking times.
3. The system must allow users to view recipe information such as ingredients, preparation steps and reviews.
4. Users can adjust the serving size of the recipe and the system must recalculate the ingredient quantities accordingly.
5. The system must allow users to shop for the meal they viewed and check the availability of the recipe ingredients from Local Suppliers.
6. The system must offer substitutes based on the ingredient taxonomy, or alert users if exact ingredients are unavailable, and generate a shopping cart with exact portions needed.
7. Users can complete checkout using a mock payment system, and the system must deduct stock from suppliers upon payment.
8. The system must record the cooking activity to update the Verified Chef's royalty metrics.
9. Home Cooks and Verified Chefs can create and publish recipes by supplying recipe information, selecting ingredients from the standardized taxonomy with base quantities, adding preparation steps and uploading media files.
10. The system should create and update the royalty dashboard for Verified Chefs based on the number of times the recipe was cooked or logged.

11. The system allows users to browse, join active challenges, track their progress based on logged or shopped qualifying recipes, and update leaderboards with assigned rewards upon completion.

2.2 Non-Functional Requirements

The following requirements define the operational qualities of the Culinary Discovery & Meal Kit Marketplace, focusing on system performance, data integrity, security, scalability, and availability.

1. **Performance**: The system must execute complex, multi-source inventory and substitution queries in under 1 second to ensure a seamless user experience (e.g. purchase conversion, scanning recipes).

2. **Data Integrity and Concurrency:** ACID properties must be strictly held during checkout, as concurrent transactions have risk of overselling. When two Home Cooks attempt to purchase the last available unit of an ingredient simultaneously, the system must permit only one transaction to proceed. If a transaction fails due to stock depletion, the system must roll back completely to maintain a consistent state.

3. **Security and Access Control:** The system should implement a Role-Based Access Control (RBAC) model to provide a distinction in data access. Users' access to data should be strictly restricted according to their roles, and large-scale access rights should be limited only to administrators.

4. **Scalability:** The system architecture should be able to support significant increases in usage that will occur during certain periods when system usage increases, such as Kitchen Challenge periods. And while supporting these increases, it should be able to continue without reducing response time in daily and core functions.

5. **Availability:** Since the system will have real-time users, it should be 99.9% accessible in order to support the transactions of these users. Planned maintenance outages to be carried out on the system should be performed outside of peak traffic hours in order to minimize the negative impact of this process on usage.

3. System Limitations

1. **No Real Payment Processing**
   The system uses a mock payment mechanism based on virtual balances stored in the database.
   The system will not integrate real world payment gateways, process real credit/debit card information, handle refunds, or ensure financial compliance standards.

2. **Basic Content Moderation and Verification**
   Verification of chefs and suppliers, as well as recipe categorization, is performed manually by administrators.
   The system will not implement advanced automated moderation such as plagiarism detection, fake review detection, or identity verification beyond platform level approval.

3. **Limited Media Management Capabilities**
   Media associated with recipes (images/videos) is stored as metadata or simple uploads.
   The system will not support advanced media processing features such as transcoding, streaming optimization, or large scale content delivery networks.

4. **Platform Dependent Challenge Tracking**
   Kitchen Challenge progress is recorded only through cooking logs and purchases made within the platform. The system will not validate cooking activities through external devices or any kind of real world verification mechanisms.

5. **No Multi Currency or Tax Jurisdiction Support**
   The system operates exclusively under TRY and Turkish tax regulations.
   The system will not support multi currency pricing, exchange rate conversions, or cross border tax rules.

6. **No Historical Price or Inventory Versioning**
   The system stores only the current price and stock level of supplier products.
   The system will not maintain historical records of prices or inventory states over time.

4. Additional Functionalities & Database Features

In order to enhance the Culinary & Meal Kit Marketplace beyond its default requirements, the system expands with three advanced, database-centric features. These additions require complex relational features of modeling E/R diagrams like unary and ternary relationships and so on while allowing us to construct advanced SQL implementations such as dynamic views or automated triggers to alarm the users.

1. Recipe Forking and Version Control

Instead of forcing users to create the recipes from scratch when they decided to change some ingredients or modify the recipe, the system supports “Recipe Forking”. Users can duplicate, alter, and publish an adapted version of an existing recipe while maintaining the roots of the recipe (where & whom the recipe come from).

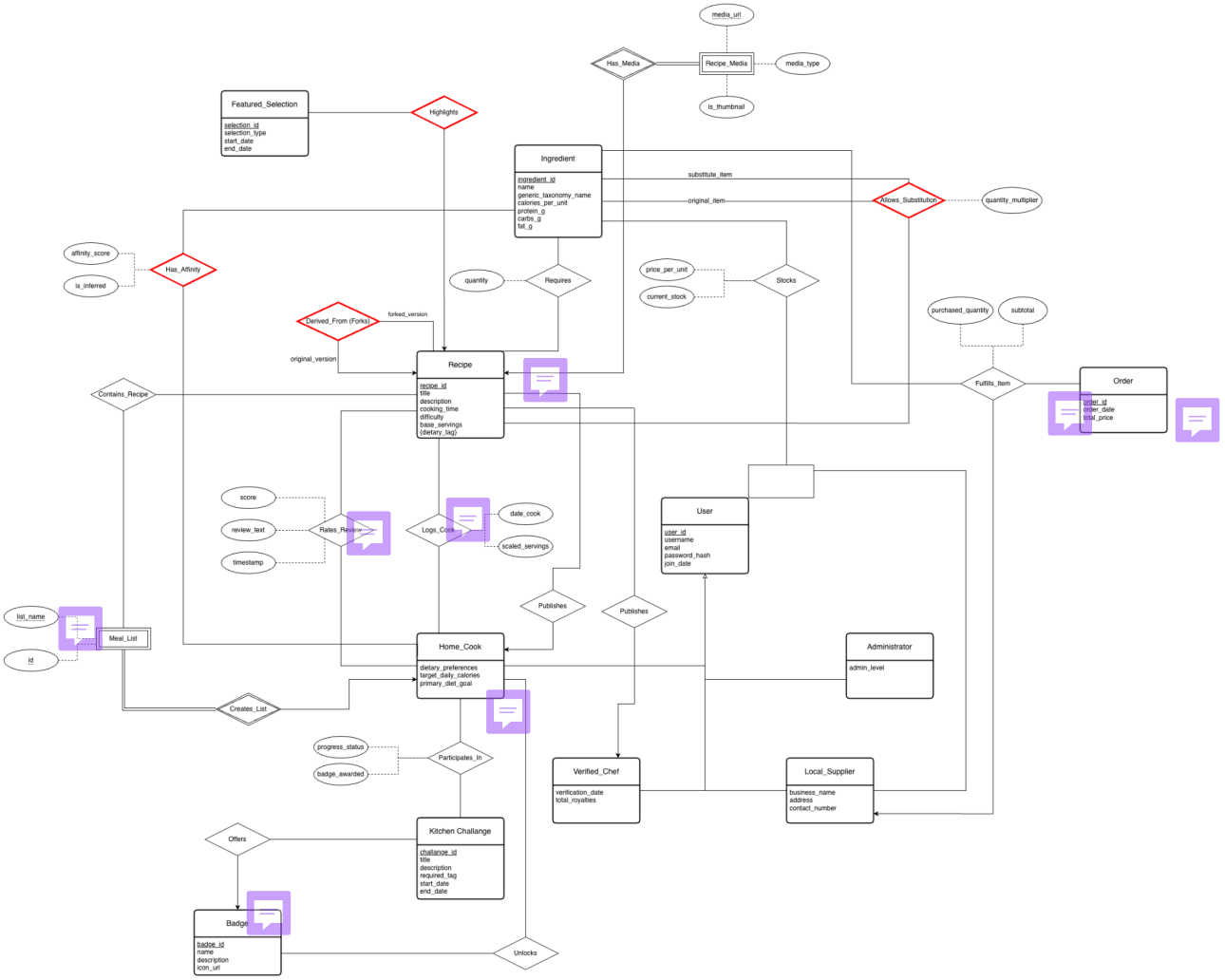
2. Context-Aware Ingredient Substitution Engine

Although ingredient substitutions are most of the time strictly dependent on the culinary context (while substituting applesauce for oil works in baking recipe, fails in pan-searing recipe etc.) If a local supplier is out of stock of a required item, the system automatically checks for a valid, recipe-specific substitute to prevent cart progress errors.

3. Hybrid Behavioral Recommendation System

The website utilizes a dual-layer recommendation engine that combines both the user’s preferences and their tracked behavioral data. It guarantees that suggested recipes mathly fit within a user’s target macros while sorting them based on the ingredients the system has learned the user most prefers.

5. E/R Diagram



*Relations shown as red represent additional features.

Here is the link for the diagram (for better readability):
[cs353project.drawio]